HIROKI KUJI PH.D. STUDENT

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EDUCATION

Department of Physics, Graduate School of Science
Tokyo University of Science, Tokyo, Japan
Ph.D. student in Physics Apr. 2025 – Present

- Supervisor: Prof. Tetsuro Nikuni

Department of Physics, Graduate School of Science
Tokyo University of Science, Tokyo, Japan
Master of Science in Physics Apr. 2023 – Mar. 2025

- Supervisor: Prof. Tetsuro Nikuni
- Thesis: "Proposal for realizing quantum-spin systems on a two-dimensional square lattice with Dzyaloshinskii–Moriya interaction by the Floquet engineering using Rydberg atoms"

Department of Physics, Faculty of Science
Tokyo University of Science, Tokyo, Japan
Bachelor of Science in Physics Apr. 2019 – Mar. 2023

RESEARCH INTERESTS

Quantum computation and quantum simulation
Quantum algorithms
Quantum error correction and error mitigation
Quantum information theory
Open quantum systems and non-Hermitian dynamics

PUBLICATIONS

Peer-reviewed articles

1. **H. Kuji**, T. Nikuni, and Y. Shingu. "Variational Quantum-Neural Hybrid Imaginary-Time Evolution." *J. Phys. Soc. Jpn.* **95**, 114003 (2026). doi:10.7566/JPSJ.95.114003.
2. R. Sato, Y. Aota, T. Yoshida, H. Kawaguchi, Y. Mori, **H. Kuji**, and Y. Matsuzaki. "Quantum circuit learning using complex spectra of non-integrable systems." *Jpn. J. Appl. Phys.* **65**, 112004 (2026). doi:10.35848/1347-4065/ae6f5b.
3. M. Tatsuta, Y. Matsuzaki, **H. Kuji**, R. Hamazaki, and A. Shimizu. "Characterization and generation of a standard-quantum-limit-beating catlike state through repetitive measurements." *Phys. Rev. A* **113**, 022412 (2026). doi:10.1103/ljb3-sdvb.
4. **H. Kuji**, M. Kunimi, and T. Nikuni. "Proposal for realizing quantum-spin systems on a two-dimensional square lattice with Dzyaloshinskii–Moriya interaction by the Floquet engineering using Rydberg atoms." *Phys. Rev. A* **112**, 022614 (2025). doi:10.1103/PhysRevA.112.022614.
5. Y. Machida, **H. Kuji**, Y. Mori, H. Kawaguchi, T. Imoto, Y. Takeuchi, M. Ishizaki, and Y. Matsuzaki. "Identification of phase plate properties using photonic quantum sensor networks." *Jpn. J. Appl. Phys.* **64**, 082002 (2025). doi:10.35848/1347-4065/adf427.
6. **H. Kuji**, Y. Shingu, T. Nikuni, T. Imoto, K. Sugisaki, and Y. Matsuzaki. "Robust phase estimation of the ground-state energy without controlled time evolution on a quantum device." *Phys. Rev. A* **111**, 042618 (2025). doi:10.1103/PhysRevA.111.042618.

Preprint

7. **H. Kuji**, S. Endo, T. Nikuni, R. Hamazaki, and Y. Matsuzaki. "Quantum Error Mitigation Simulates General Non-Hermitian Dynamics." arXiv:2602.21879 (2026).

Fellowships and Research Experience

JSPS Research Fellow (DC2)
Japan Society for the Promotion of Science — Apr. 2026 – Mar. 2028

- Research project: Development of methods for simulating non-Hermitian dynamics on quantum computers (Grant No. 26KJ2001).

Visiting Researcher
University of Stuttgart, Germany — Dec. 2025 – Oct. 2027

- JST ASPIRE Japan–Germany project: "Quantum computing with neutral atoms".
- Institute for Theoretical Physics III; Prof. Hans-Peter Büchler's group.

Research Assistant — Prof. Tetsuro Nikuni's group
Tokyo University of Science, Tokyo, Japan — Apr. 2025 – Nov. 2025

- Project: "Theoretical research on the development and applications of neutral-atom quantum computers and quantum simulators".

Research Assistant — Prof. Yuichiro Matsuzaki's group
Chuo University, Tokyo, Japan — Aug. 2024 – Nov. 2025

- JST Moonshot Goal 6 project: "Scalable and Robust Integrated Quantum Communication System".

Research Assistant — Prof. Takahiko Satoh's group
Keio University, Kanagawa, Japan — Apr. 2024 – May 2025

- NEDO project: "Quantum circuit partitioning library", under "Development of Quantum-Classical Hybrid Use-Case Technologies in Cyber-Physical Space".

Teaching Experience

Teaching Assistant
Tokyo University of Science, Tokyo, Japan — Apr. 2023 – Mar. 2025

- Assisted in undergraduate courses including Quantum Mechanics IB, Electromagnetism I, General Physics, and Mathematics for Physics.
- Supported exercises across the Physics and Economics departments; led problem-solving sessions, prepared worksheets and solutions, and supported grading and office-hour Q&A.

Selected Presentations

STAR architecture with erasure conversion
JP-DE ASPIRE "Neutral-atom QC", Tübingen, Germany (Invited talk) — Sep. 2026

Quantum Error Mitigation Simulates General Non-Hermitian Dynamics
JP-DE ASPIRE "Neutral-atom QC", Tübingen, Germany (Poster) — Sep. 2026

Robust phase estimation of the ground-state energy without controlled time evolution on a quantum device
QIP 2026, Latvia (Poster) — Jan. 2026

Robust phase estimation of the ground-state energy without controlled time evolution on a quantum device
2nd Int. Workshop on Quantum Information Engineering, Japan (Oral) — Oct. 2025

Proposal for realizing quantum-spin systems on a two-dimensional square lattice with Dzyaloshinskii–Moriya interaction by the Floquet engineering using Rydberg atoms
BEC 2025, Spain (Poster) — Sep. 2025

Robust phase estimation of the ground-state energy without controlled time evolution on a quantum device
AQIS 2025, Hong Kong (Poster) — Aug. 2025

Variational quantum-neural hybrid imaginary time evolution
Quantum Innovation 2025, Japan (Poster) — Jul. 2025

Proposal for realizing quantum-spin systems on a two-dimensional square lattice with Dzyaloshinskii–Moriya interaction by the Floquet engineering using Rydberg atoms
Kindai QPT Seminar, Kindai University, Japan (Invited seminar) — May 2025

Proposal for realizing quantum-spin systems on a two-dimensional square lattice with Dzyaloshinskii–Moriya interaction by the Floquet engineering using Rydberg atoms
APS Global Physics Summit, USA (Oral) — Mar. 2025